Lingzhi Zhao

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RESEARCH INTERESTS

My research focuses on **ML-based end-to-end video systems**, spanning **video codecs**, **real-time streaming**, **adaptive bitrate**, and **video quality assessment**.

EDUCATION

University of Illinois Urbana-Champaign · Ph.D. in Computer Science — Aug. 2022 – May 2027
Advisor: Prof. Klara Nahrstedt

Shanghai Jiao Tong University · M.S. in Electrical and Computer Engineering — 2019 – 2022

Shanghai University · B.S. in Electrical and Computer Engineering — 2015 – 2019

TECHNICAL SKILLS

Video Systems & Tools	ABR, rate control, DASH, VQA, FFmpeg, NVCodec, Wireshark, iperf3, tc, NS-3
Programming	Python, C++, C#, Swift, Java, Rust, Git
Machine Learning	PyTorch, Transformers, VLM/MLLM, SFT, RL, on-device inference

RESEARCH & WORK EXPERIENCE

University of Illinois Urbana-Champaign, MONET Lab — Aug. 2022 – Present

Graduate Research Assistant · Advisor: Prof. Klara Nahrstedt

GenStream: Generative Codecs for Real-Time Video Streaming

- Optimized a **generative video codec** (Cosmos Tokenizer) for **real-time streaming** on a single RTX 4090, achieving **50%–78% bandwidth savings** over H.264/H.265 and prior neural codecs at comparable quality.
- Designed **content-aware token adaptation** for changing network conditions, improving perceptual quality (LPIPS) by **38%–59%** at matched bitrates and loss patterns versus state-of-the-art streaming systems.

AquaScope: Generative Codecs for Mobile Image Transmission *[ACM MobiSys 2026] [Website]*

- Optimized visual tokenizers (TiTok, VQGAN) through **token clustering and fine-tuning** for **low-bandwidth, error-resilient image transmission**.
- Integrated the learned codec into a mobile acoustic prototype, improving **bandwidth efficiency** by **40%** and reducing average **BER** from **19% to below 2%**; outperformed PNG, JPEG, and classic neural compression.

AquaVLM: VLM-powered Intelligent Mobile Communication Systems *[ACM UbiComp 2027] [Website]*

- Built an **end-to-end interactive VR application** for underwater communication, using **on-device VLM inference** over **multimodal inputs** to enable divers to generate and exchange context-aware messages.
- Fine-tuned MobileVLM-V2 (3B) via SFT on an 85K-sample multimodal dataset constructed with GPT-4o few-shot prompting to support **context-aware message generation**.
- Designed a **hierarchical message-generation** pipeline, achieving **>80% user-intent alignment** while reducing inference latency by **60%–70%**.
- Deployed the VLM on **iOS mobile devices** using 4-bit quantization with LLMFarm for an **interactive underwater communication prototype**.

ByteDance Inc., San Diego, CA — May 2024 – Aug. 2025

Research Intern, Multimedia Lab · Manager: Dr. Shu Shi

MARS: Motion-Aware ABR for Real-Time VR *[Submitted to ACM MMSys 2027]*

- Developed **perceptually guided bitrate adaptation for real-time VR streaming**, reducing bandwidth usage by **15%** without degrading quality of experience (QoE), validated in a user study with diverse VR applications.
- Designed a **kinematics-based state machine** to synchronize bitrate adaptation with **head-motion phases**; integrated motion-vector information and support for CBR, VBR, and CQP rate control for the system.

Trinity: Low-Latency Cloud VR Gaming *[ACM MMSys 2026] [Website]*

- Prototyped an **end-to-end cloud VR gaming system**, reducing bandwidth cost by **60%** and GPU utilization by **40%** while maintaining QoE **comparable to local streaming**.
- Characterized **latency tolerance across head, hand, and body motion** using an open-source first-person shooter VR game.
- Designed **delayed rendering for hand motion** and **partial-panorama rendering for head motion** to account for action-dependent latency sensitivity.

TikTok, San Jose, CA May 2026 – Present

Machine Learning Engineer Intern · Manager: Prof. Feng Qian

Video Quality Assessment for Short-Video Transcoding *[Submitted to ICLR 2027]*

- Built a **short-video transcoding quality benchmark** with **377K subjective ratings**, spanning source resolutions, transcoding settings, and mobile devices; evaluated MLLM scorers and analyzed their failure modes.
- Post-trained a **context-aware MLLM with motion–artifact fusion** via SFT, achieving state-of-the-art quality prediction on the new benchmark and three public datasets.

Dynamic Ad Object Insertion for UGC Videos *[Manuscript in preparation]*

- Built an end-to-end pipeline for **dynamic ad object insertion**, combining video matching, surface detection, and compositing with latency comparable to video transcoding, serving as a proof of concept for online deployment.
- Accelerated SAM3 surface detection by **8.5**× via **adaptive frame sampling** and fine-tuned a Qwen3-VL embedding model for **brand-conflict detection and contextual matching**.

SELECTED PUBLICATIONS

[1] **Lingzhi Zhao**, Yu Liu, Yilin Li, Diane Xu, Shi Han Chong, Yang Fan Mia, Xinyi Liu, Yufei Li, Xinheng Sheng, Isaac Zhang, Chen Zheng, Zhengyuan Yang, Lei Shi, and Feng Qian. Short UGC Video Quality Assessment via Large Language Model: Benchmark and Method. ***Submitted to ICLR 2027.***

[2] **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Mingyuan Wu, Bo Chen, Ruixiao Zhang, Klara Nahrstedt, and Shu Shi. MARS: Streaming How You Move with Motion-Aware ABR for Real-Time VR. ***Submitted to ACM MMSys 2027.***

[3] Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Haozhen Zheng, Jingcheng Yang, Mingyuan Wu, Deepak Vasisht, and Klara Nahrstedt. AquaVLM: Improving Underwater Situation Awareness with Mobile Vision Language Models. (*: equal contributions) ***ACM UbiComp 2027.*** *(corresponding author)*

[4] Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Mingyuan Wu, Haozhen Zheng, Deepak Vasisht, Francis Y. Yan, and Klara Nahrstedt. AquaScope: Reliable Underwater Image Transmission on Mobile Devices. (*: equal contributions) ***ACM MobiSys 2026.*** *(corresponding author)*

[5] **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Ruixiao Zhang, Shu Shi, and Klara Nahrstedt. Trinity: Exploiting Latency Sensitivity to Improve Quality of Experience on Cloud VR Gaming. ***ACM MMSys 2026.***

[6] **Lingzhi Zhao**, Ying Cui, Yuhang Jia, Yunfei Zhang, and Klara Nahrstedt. Enhancing Neural Adaptive Wireless Video Streaming via Lower-Layer Information Exposure and Online Tuning. ***IEEE Transactions on Multimedia 2025.***

[7] **Lingzhi Zhao**, Qian Zhou, Bo Chen, and Klara Nahrstedt. 360LiveCast: A Low-Latency and Bandwidth-efficient Multicast Framework for Live 360 Video. ***IEEE MIPR 2025.***

[8] **Lingzhi Zhao**, Ying Cui, Sheng Yang, and Shlomo Shamai (Shitz). An Optimization Framework for General Rate Splitting for General Multicast. ***IEEE Transactions on Wireless Communications 2022.***

[9] **Lingzhi Zhao**, Ying Cui, Zhi Liu, Yunfei Zhang, and Sheng Yang. Adaptive Streaming of 360 Videos with Perfect, Imperfect, and Unknown FoV Viewing Probabilities in Wireless Networks. ***IEEE Transactions on Image Processing 2021.***